Assignment 1

Objective:

- Use an LDR (Light Dependent Resistor) to control an LED based on ambient light levels by reading the LDR value through an ADC pin on the microcontroller.

Components:

- ATmega32 microcontroller
- LDR (Light Dependent Resistor)
- LED
- Resistors
- wires

Circuit Setup:

- Connect the LDR in a voltage divider configuration with a fixed resistor, with the midpoint connected to an ADC pin on the ATmega32.
- Connect the LED to a digital output pin on the microcontroller with a current-limiting resistor.

Programming:

- Initialize the ADC to read analog values from the LDR.
- Set a threshold value to determine when the LED should turn on or off based on the LDR reading.
- Continuously read the LDR value and compare it to the threshold.
- Turn the LED on if the light level is below the threshold (indicating darkness) and off if above the threshold (indicating brightness).

Operation :

- The LED should turn on when the environment is dark (low LDR reading) and turn off when it is bright (high LDR reading).